

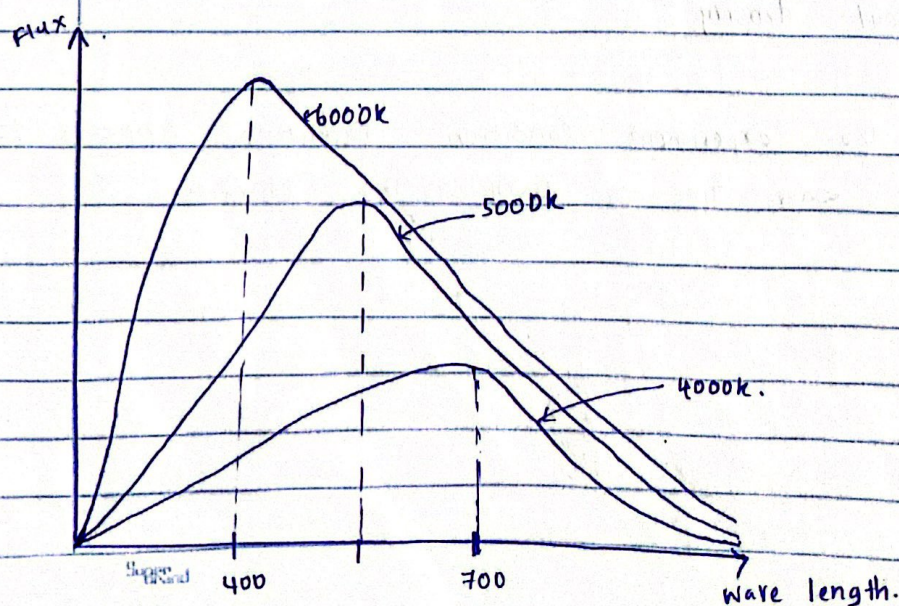
Black body radiation

Black body radiation refers to the electromagnetic radiation that is emitted from a perfect black body at a room temperature. A black body is an idealized object that absorbs all the radiation incident upon it and re-emits it in a continuous spectrum. The radiation emitted by a black body is dependent only on its temperature with the spectral distribution of the radiation following plank's law.

The term 'black body' refers to an object that is perfectly black, which means that it completely absorbs all radiation is important in understanding the thermal properties of materials, the structure of stars, and the development of theory of quantum mechanism.

Q. What is ultra violet catastrophe?

It was predicted that as the frequency of the radiation, increased, the energy radiated by a black body would become $\propto$ infinitely large.



Date:.....

Rayleigh - Jeans law

$$u(\nu, T) = \frac{8\pi k_B T}{c^3} \nu^2 d\nu$$

k_B - Boltzmann constant

c - velocity.

density of radiation energy
between ν ($\nu + d\nu$) and T

This law is agreed experiment result,
at very low frequency (ν)
disagree with at higher frequency (ν)

* This law is only valid low frequency.

~~The Rayleigh - Jeans law accurately describes the behaviour of blackbody radiation at low~~

The Rayleigh Jeans law is an approximation that describes the spectral radiance of electromagnetic radiation emitted by a black body at thermal equilibrium.

The Rayleigh Jeans law accurately describes the behaviour of black body radiation at low frequencies. but fails to match experimental observations at high frequencies. This is due to the ultra violet catastrophe, as predicted by the Rayleigh ~~Jeans~~ law, where the intensity of radiation would increase without bound as frequency increases.

The failure of the Rayleigh Jeans law to explain the observed behaviour of black body radiation at high frequency was resolved by Max Planck's introduction of quantum theory and the concept of energy quantization.

Signature

No:.....

Plank's law successfully described the entire spectrum of black body radiation, including the high and low frequency regions. by treating energy as discrete packets or photons.

Plank's Law

Plank's law provides an accurate description of the distribution of energy emitted by a black body at a given temperature across a range of frequencies or wave length.

Plank's law successfully resolves the ultraviolet catastrophe predicted by Rayleigh Jeans law.

His assumption

He assume that frequency of the electronic oscillator must be a minimum energy (E) or an integrated multiple of energy.

r - Spectral radiance at given wave length

T - Temperature.

c - Speed of the light.

$$P(r, T) dr = \frac{8\pi h r^3}{c^3} \cdot \frac{1}{\exp\left(\frac{hr}{k_b T}\right) - 1}$$

$$h = 6.63 \times 10^{-34} \text{ Js.}$$

Plank's quantum hypothesis states that energy is quantized, it means, it is emitted and absorbed in discrete packets or bundles of energy called photons.

* photoelectric effect was explained by Elbert ~~Blum~~ Einstein.

Date:

Black body radiation and the photoelectric effect

Photoelectric effect is a phenomenon where electrons are emitted from a material surface when it's illuminated with light or electro magnetic radiation.

while both black body radiation and photo electric effect involves the behaviour of electromagnetic radiation, they are distinct phenomena with different underlying principles.

2nd day

Balmer Formula (1885)

Relates the frequencies (or wave length) of spectral lines in the hydrogen's atom's emission spectrum. The blamer formula is specially applicable to the visible region of the spectrum electromagnetic spectrum.

$$\bar{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (n_2 > n_1)$$

$n = 1, 2, 3, 4, \dots$

↑

quantization of energy level.

$$\bar{\nu} = \frac{1}{\lambda}$$

$$\boxed{\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)}$$

λ - wave length of the spectral line.

R_H - Rodberg constant.

or
(ആറ്റി തിരയെ തിര നീളം)

← note.

Niels Bohr → Idea of quantization of energy level.

↓ It explain

emission spectrum of H atom.

Bohr's model was primarily concerned with explaining the behaviour of electrons within atoms. He proposed that electron occupy specific, quantized energy levels or orbits around the nucleus. Each energy level corresponds to a specific amount of energy that the electron possesses.

According to the Bohr's model electrons can transition between these energy levels by either absorbing or emitting energy in discrete amounts. When an electron absorb energy, it moves to a higher energy level, or when it emits energy, it drops to a lower energy level. The energy transition occur in photons, with the energy of each quantum determined by the difference in energy between the initial and final energy levels.

Rutherford Explanation

Rutherford proposed that atoms have a tiny, dense and positively charged nucleus at the center. The majority of the atom's mass is concentrated in this nucleus, while the negatively charged electrons orbit around the nucleus at a distance.

According to electro magnetic radiation, Charge particle moves ⁱⁿ a circular path. and it should emit energy.

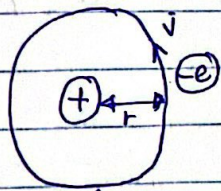
Bohr model of the atom

Bohr use the plank's quantum theory the explain the line spectrum of H.

His assumption / postulation

(1) Negative charge electrons are evolving it's circular orbits known as stationary energy levels and electron allowed stationary energy states will not spiral in to the nucleus.

(2) Angular momentum of the electron is quantize by $h/2\pi$.

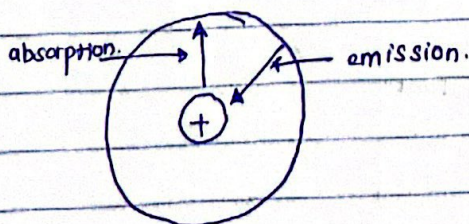


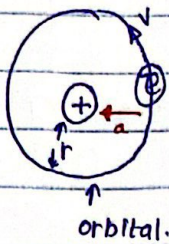
↑ stationary energy level.

$$mvr = \left(\frac{h}{2\pi} \right) n \quad n = 1, 2, 3, 4, \dots$$

(3) Electron does not emit radiation as long as it is in a fixed orbital or energy level.

(4) Electron emit or absorb energy only when they to transfer between two energy level.





$$\text{centripetal force} = \frac{mv^2}{r}$$

Electro static attraction $\rightarrow$ because of electron is negative and nucleus is positive.

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

r - distance between the two particles.

$\frac{1}{4\pi\epsilon_0}$ - dielectric permittivity of the free space.

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}$$

$$\text{but } mvr = n \left(\frac{h}{2\pi} \right)$$

$$v = \frac{nh}{2\pi mr}$$

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m n^2 h^2}{4\pi^2 m^2 r^3} \quad \text{--- (2)}$$

$$r = \frac{\epsilon_0 n^2 h^2}{\pi m e^2} \quad \leftarrow \text{constant}$$

$$r = n^2 \cdot \text{constant}$$

eg = 1st orbit $n=1$ 2nd orbit $n=2$

$$r = \frac{n^2 \epsilon_0 h^2}{\pi m e^2} \text{ constant}$$

$$r = n^2 \text{ constant}$$

$$r = 2^2 \times 5.29 \times 10^{-11}$$

$$r = 4 \times 5.29 \times 10^{-11} \text{ m}$$

$$r = 1 \text{ constant}$$

$$r = 1 \times 5.29 \times 10^{-11} \text{ m}$$

Energy

$$\text{Total Energy (E}_T\text{)} = \text{kinetic energy (K.E)} + \text{Potential energy (P.E)}$$

$$= \frac{1}{2} m v^2 + \frac{-e^2}{4\pi\epsilon_0 r} \quad \text{--- (1)}$$

From equation (2).

$$\frac{m v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2} - m v^2 = \frac{e^2}{4\pi\epsilon_0 r}$$

$$E_T = \frac{1/2 e^2}{4\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r}$$

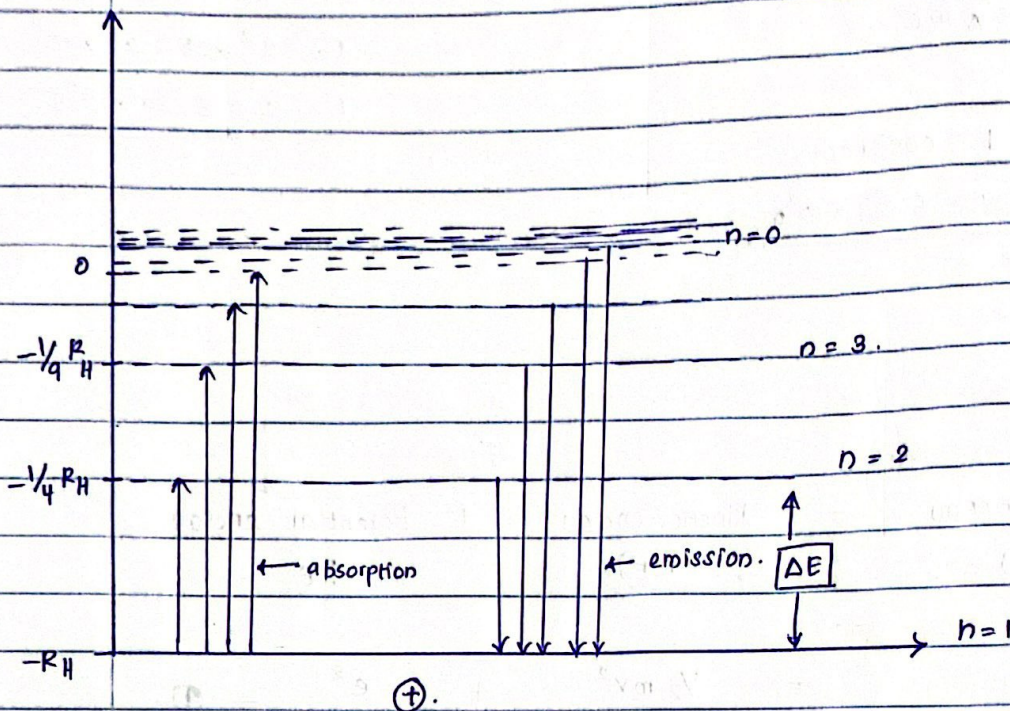
$$E_T = \frac{-e^2}{8\pi\epsilon_0} \times \frac{\pi m e^2}{\epsilon_0 n^2 h^2}$$

$$E_T = - \frac{m e^4}{8 \epsilon_0^2 n^2 h^2} \leftarrow \text{constant}$$

$$E_T = - \frac{1}{n^2} \times \text{constant}$$

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The negative sign of the orbit energy indicates less energy compare to free electron.



$$E_2 = \frac{-me^4}{8\pi\epsilon_0^2 n_2^2 h^2}$$

$$E_1 = \frac{-me^4}{8\pi\epsilon_0^2 n_1^2 h^2}$$

$$\Delta E = E_2 - E_1$$

$$= \frac{-me^4}{8\pi\epsilon_0^2 n_2^2 h^2} - \left(\frac{-me^4}{8\pi\epsilon_0^2 n_1^2 h^2} \right)$$

$$\Delta E = \frac{me^4}{8\pi\epsilon_0 h^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\gamma = c/\lambda$$

$$\Delta E = h\gamma$$

$$= h \frac{c}{\lambda}$$

$$\Delta E = hc \frac{1}{\lambda}$$

$$\lambda = \Delta E/hc$$

$$\bar{r} = \frac{me^4}{8\epsilon_0 h^3 c} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\bar{r} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$E \propto r$$

$$E \propto \frac{1}{\lambda}$$

Wilson introduce another quantum number (l)

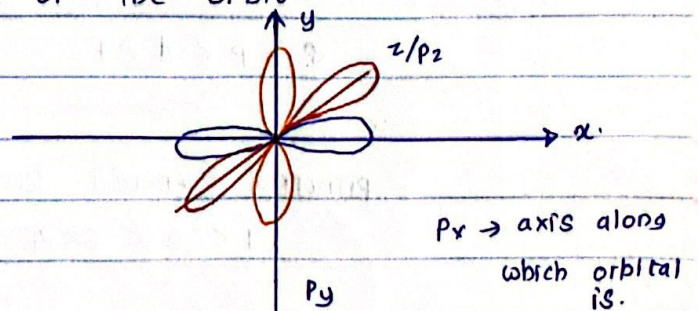
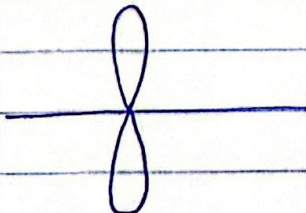
l = azimuthal quantum number.

$$l = n - 1$$

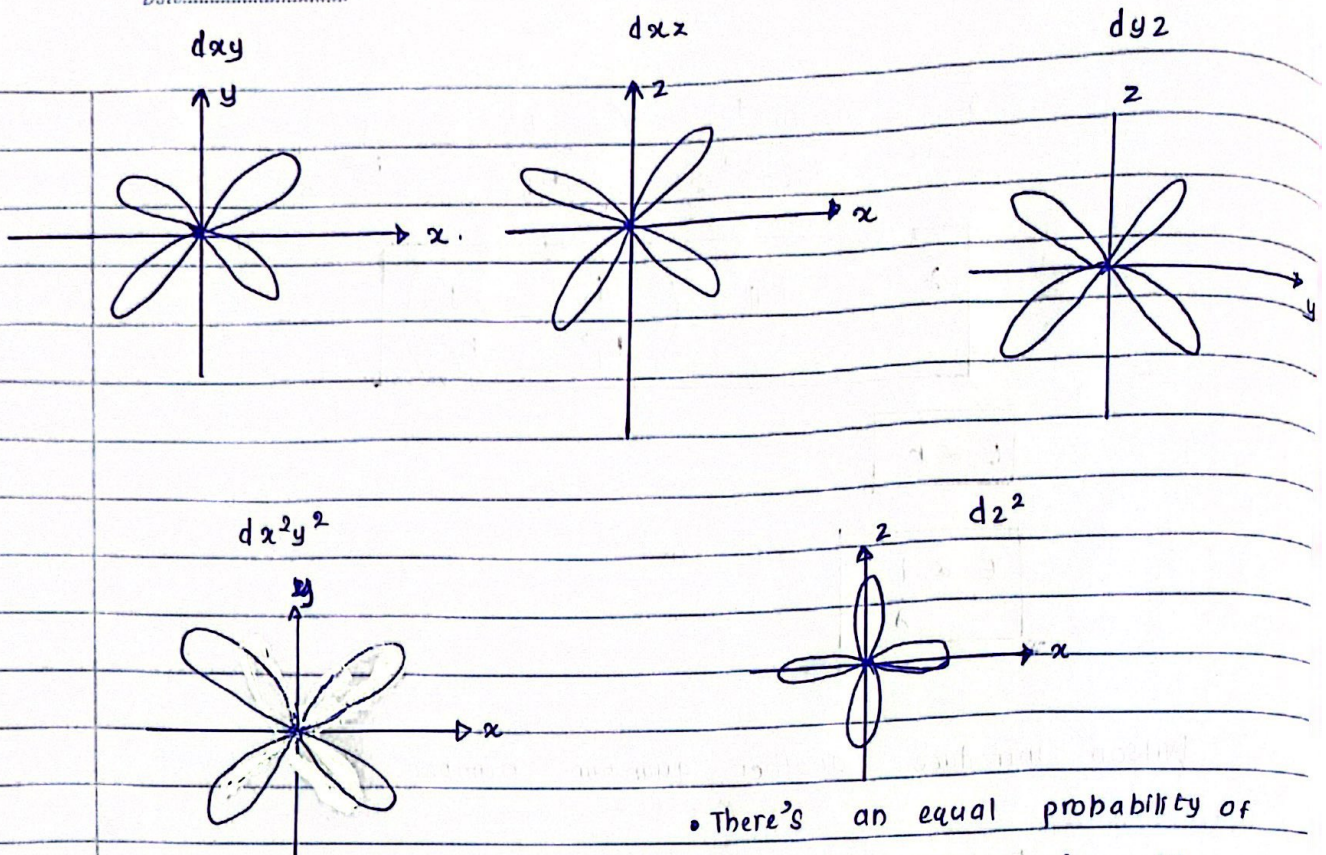
eg:- $n=1$, $l=0$
 $n=2$, $l=1, 0$
 $n=3$, $l=2, 1, 0$.

This describes the angular momentum of the electron, relative to the nucleus.

This also determines the shape of the orbit.



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• There's an equal probability of being an electron in either lobe p orbital.

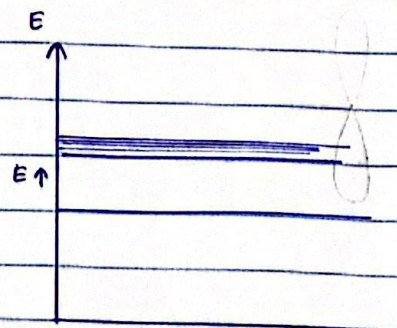
$n = 3$ energy level has 5 d orbitals.

$n = 4$ energy level has 7 d orbitals.

If we look at the order of the energy of sub energy levels for a specific principal quantum number it increases s orbitals to f orbitals.

$$s < p < d < f$$

principal energy level
 $1 < 2 < 3 < 4$

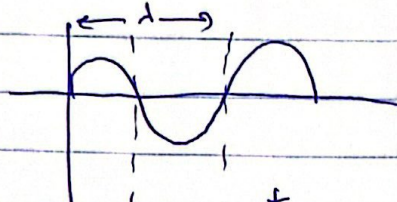


Wave particle duality

microscopic objects show both wave & particle properties.

Interferences

De brogli

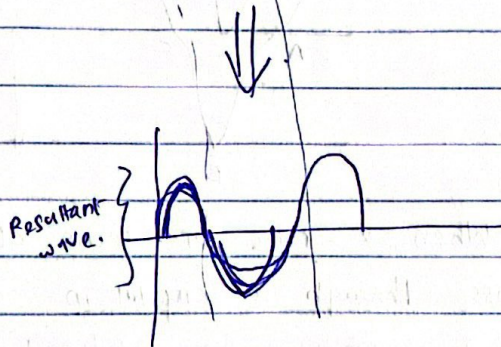


$$E = hf = hc/\lambda$$

Two waves in

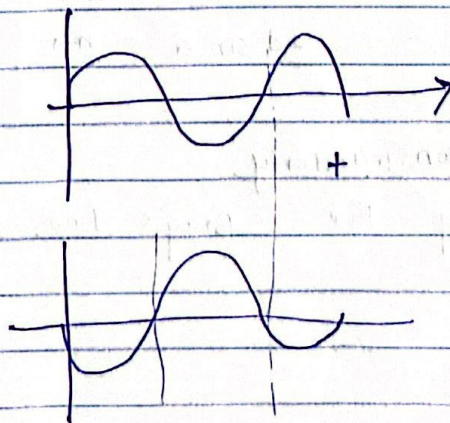
Evidences show that electrons behave as waves.

- Interference.
- Diffraction.
- Reflection.



* The energy does not change. but have higher altitude.

Both light and electron show this feature.



Resultant wave

destructive Interferences

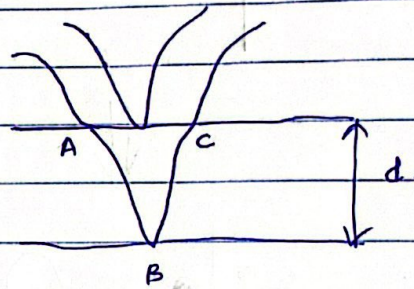
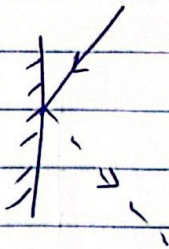
No:

(II) Diffraction.

Spreading of waves after they pass through small openings.



(III) Reflections.



When x-rays or beam of electrons pass through a crystal in different angle where the pathway $(AB + BC)$ difference is an integral multiple of wave length.

$$AB + BC = n\lambda$$

$$2d \sin \theta = n\lambda$$

Interfere constructively,
satisfy the Bragg's law.

Evidences to show the particle nature of electrons.

(01.) cathode ray tube experiment.

• The wheel was rotated.

∴ electrons can exert a mechanical force.

cathode rays have a momentum.

(02) photo electric effect.



The light hits the atoms.

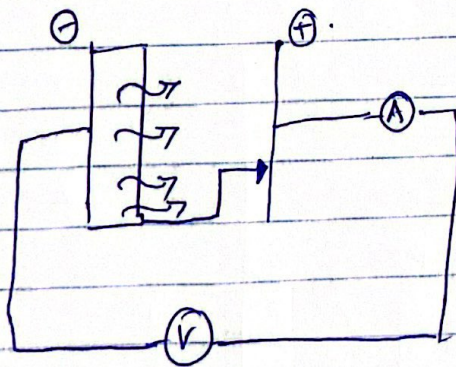
The shine can eject electrons.

(Ejection of electron).

When light was fallen
to a metal surface,
electrons were emitted.

photons of radiation with sufficient
energy strike with matter, electrons
are emitted.

This phenomenon is called
photo electric effect.



* when light is allowed to fall
on one ~~electrons~~ electrode,
photoelectrons are collected at
another electrode connected to
a positive potential of a
voltage source.

As a result a current ~~force~~ flows through a circuit which
can be measured.

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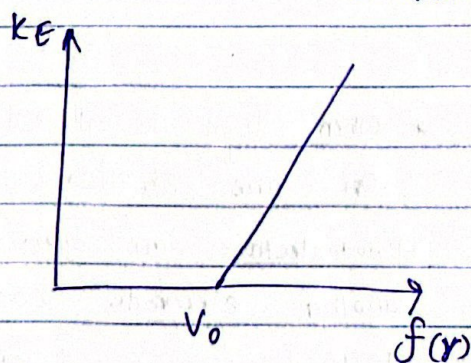
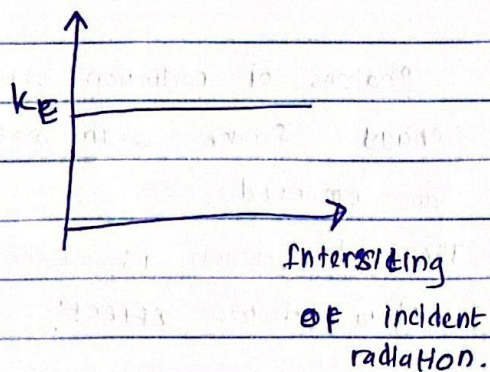
B^g applying a negative potential to the 2nd electrode photoelectrons feel repulsion negative kinetic energy of moving electrons decreases.

At a sufficiently high negative potential the motion of electrons must stop.

This potential is stopping potential.

- The energy of an electron (eV) = $\frac{1}{2}mv^2$

$$eV_s = \frac{1}{2}mv^2 \quad \text{--- (1)}$$



Assumptions

1. photoelectrons are generated ~~at~~ instantly upon irradiation, with appropriate frequency of radiation.
2. Number of photo electrons generated depends on the intensity of radiation.
3. There is a minimum frequency of incident radiation called "threshold frequency" (ν_0) below which electrons are not ejected from the metal surface.

(I) Instantaneous ^{cross} emission of photoelectrons.

(II) existence of ν_0 (threshold frequency).

(I), (II) cannot be answered by classical electromagnetic theory

— Planck's quantum theory of light —

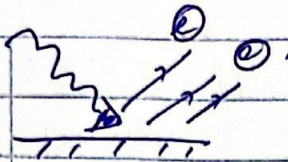
* Albert E. made a crucial statement using plank's theory.
(light is not continuous).

Light is composed discrete,

→ packets of energy called quanta.

↓
photons ← G.N. Lewis.

$$E = hf$$



$E = h\nu \Rightarrow$ Part of energy is used to overcome attractions between electrons and metal lattice.

Wave of incident form, must have minimum energy to overcome the binding energy holding the electrons. This is called, work function. (ϕ)

$$\phi = h\nu_0$$

The rest of the energy is converted to kinetic energy of electrons. (KE)

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

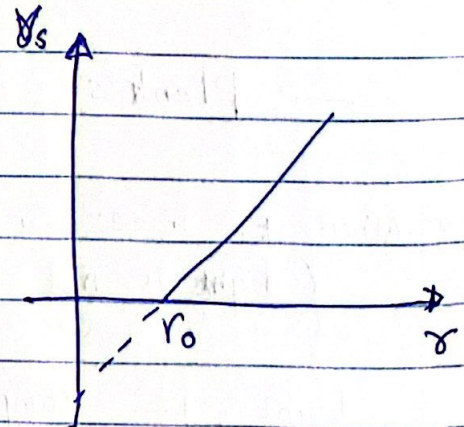
(Total Energy) (work function) (kinetic energy)

$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

$$-eV_s = h\nu - h\nu_0$$

$$V_s = -\left(\frac{h}{e}\right)\nu + \left(\frac{h\nu_0}{e}\right)$$

$\downarrow$ $\downarrow$ $\downarrow$
 y $-h$ x e



Date:.....

$$\rightarrow 4.2 \text{ m/s}$$



$$3 \times 10^8 \text{ m/s}$$

$$9.2 \times 10^{-34} \text{ kg}$$

De Broglie

$$\lambda = \frac{h}{mv}$$

$$\lambda_2 = \frac{6.626 \times 10^{-34}}{9.2 \times 10^{-34} \times 3 \times 10^8 \text{ m/s}}$$

$$\lambda_1 = \frac{6.626 \times 10^{-34} \text{ Js}}{1.57 \times 10^{-3} \text{ kg} \times 4.2 \text{ m/s}}$$

$$\lambda_2 = 4 \times 10^{-10} \text{ m}$$

$$\lambda_1 = 1.00 \times 10^{-34} \text{ m}$$

The radius of the hall is 4 cm.

Wave length is 10^{-34} m .

Wave length is very smaller compares to the size of the object.

$\therefore$ don't act as a wave.

electron wave length is compatible to the size of electron.

$\therefore$ act as a wave.

4th DayDe Broglie model of the atom

$$\lambda = \frac{h}{mv}$$

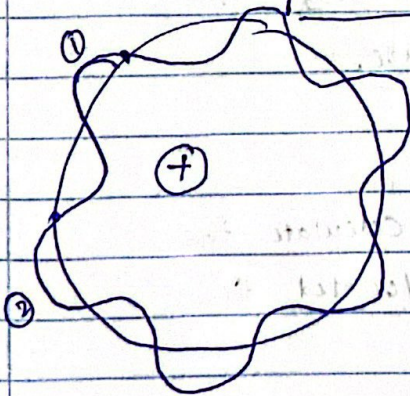
electron exist the various orbits as a ~~standard~~ standing waves. In order to fit the wave length to ^{bohr} atom the length of the orbit ($2\pi r$) should be equal to whole number multiple of the wave.

$$\text{length of the orbit } (2\pi r) = n\lambda \quad n = 1, 2, 3, \dots$$

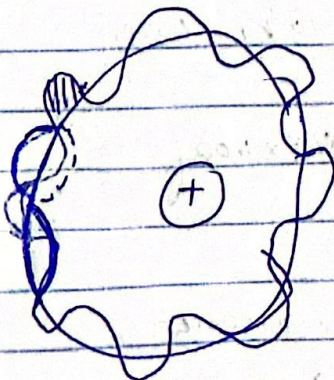
$$2\pi r = n \times \frac{h}{mv}$$

$$r = \frac{nh}{2\pi mv}$$

← agreement with
bohr's quantization
hypothesis.



constructing
filling the
orbit.



not constructing
filling the
orbit.

Heisenberg ^{uncertainty} principle

He describes fundamental limits of electron.

It is different to

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

The exact position of electron moment cannot be determine simultaneously with a desire accuracy.

$\Delta x \Rightarrow$ uncertainty in the position of the electron.

$\Delta p \Rightarrow$ uncertainty moment of the electron.

If one quantities measured accurately, the other quantity becomes less accurate.

IF we know then position $\Delta x = 0$ accurately $\Delta p = \infty$
 IF we know then moment $\Delta p = 0$ accurately $\Delta x = \infty$

$V = 600 \text{ m s}^{-1}$, accuracy of 0.005%. Calculate the certainty with which we can located the position of the electron.

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

$$mv = 9.1 \times 10^{-31} \text{ kg} \cdot 600 \text{ m s}^{-1}$$

$$\Delta p = 9.1 \times 10^{-31} \times 600 \left(\frac{0.005}{100} \right)$$

$$\Delta x = \frac{6.6 \times 10^{-34}}{4\pi} \times \frac{1}{\Delta p}$$

$$= 1.9 \text{ mm}$$

cannot be measured less than 1.99 mm

probability density

wave function

 $\psi \rightarrow$ has some energy.

It is not possible to determine the exact location of an electron in an atom. Therefore position of an electron is described in terms of probability. The probability of finding an electron given position can be calculated.

Three dimensional volume in which the probability of an electron is most likely to be is high is called orbital.

quantum
mechanics.

Erwin Schrodinger - 1926

$$\hat{H}\psi = E\psi$$

time independent.

Hamiltonian operator $\downarrow$ Total energy (PE + KE)

wave function.

energy value

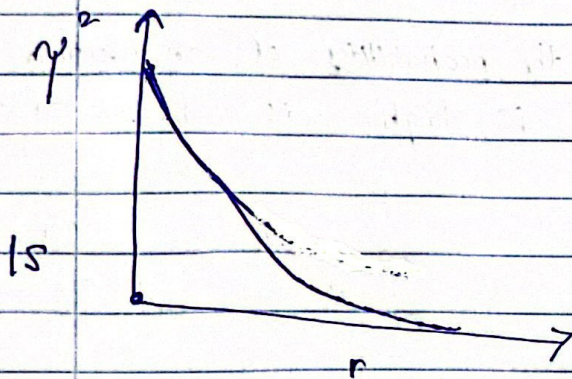
microscopic systems

- dynamics of any can be described

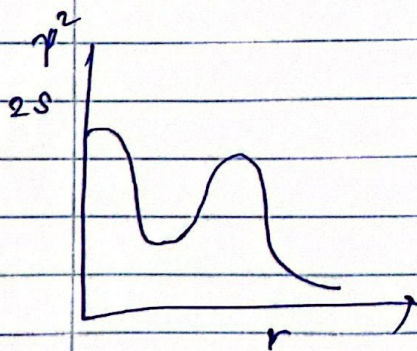
ψ has no physical meaning.

$$|\psi^2 dx dy dz|$$

The quantity of $\psi^2 dx dy dz$,
is direct measure of the probability of finding
the electron. The volume dv , is an energy E

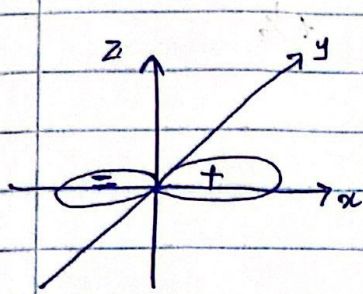


probability of finding electron all 3 directions
equally symmetrical.

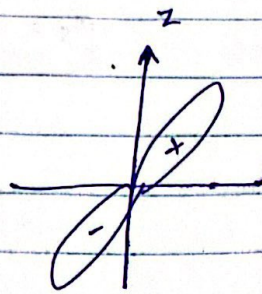


$l=1$ p orbital

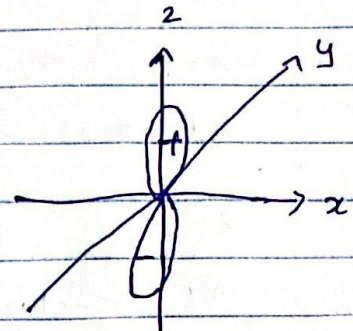
$l = 1$ p orbital.



$2p_x$.



$2p_y$



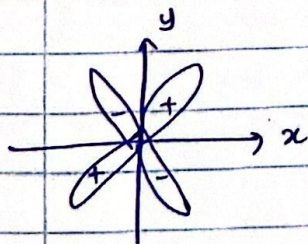
$2p_z$.

$l =$ azimuthal quantum number.

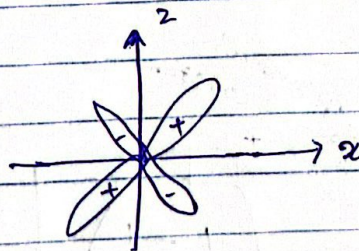
* angular momentum of the electron depending on the position of the e^- relative to the nucleus.

* determines the shape of orbit.

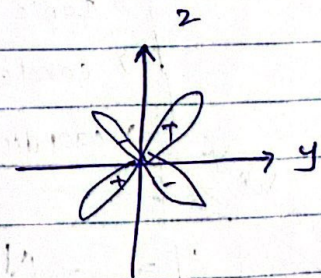
$l = 2$ d-orbitals.



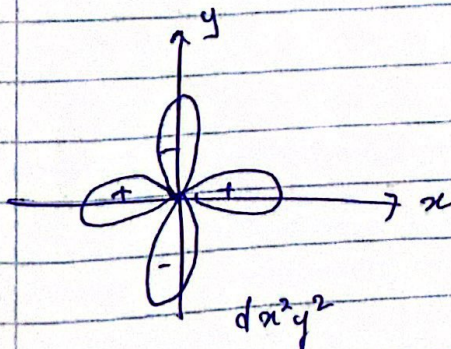
d_{xy}



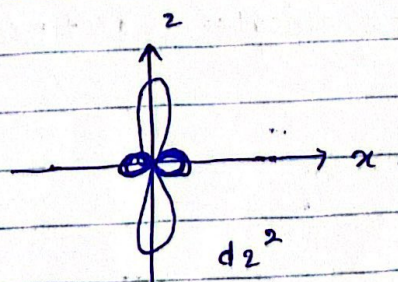
$d_{x^2-y^2}$



d_{yz}



$d_{x^2-y^2}$



d_{z^2}

Date:

n - Principle quantum number
 l - Azimuthal quantum number.
 m_l - magnetic quantum number.
 m_s - spin quantum number.

$$l = 1$$

$$m_s = -\frac{1}{2}, \pm\frac{1}{2}$$

$$m_l = -1, 0, 1$$

$$l = 2$$

$$m_l = -2, -1, 0, 1, 2$$

Pauli's exclusion principle.

• No two electrons in an atom.

can have the same set of 4 quantum numbers.

n, l, m_l, m_s .

Bonding

→ Ionic
→ covalent
→ coordinating.

Born Lande
lattice energy.

$$U = - \frac{N_A A z^2 e^2}{4 \pi \epsilon_0 r} \left(1 - \frac{1}{n} \right)$$

→ constant.

r - intermolecular distance.